The Effect of Policy Implementation on the Use of Public Transit

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1 Introduction

The notion of public transportation has been around for hundreds of years; the first concept of public transit dates back to the 1820s with the omnibus– a horse drawn car equipped to carry 10 people at a time [7]. With the growing popularity of private ride sharing companies and personal vehicles, work has been done to increase the attractiveness of public transportation. Emphasis is mainly on the speed and convenience of public transit in order to incentivise commuters to take these public services in lieu of private automobiles. Much work has been done in this field as the competition between public transit and private car increases. In the 1970s, tools for route optimization and schedule design were introduced. Sharp's [13] algorithm in 1973 focused on minimizing passenger travel time and operating costs while simultaneously meeting travel demands. Sharp's heuristic program, tested on a bus network located in Georgia, revealed a 5% reduction in travel time, while generating a 9% increase in rider participation. In 1984 that Axhausen and Smith [10] were able to form objective functions and introduce parameters like commuter demand, proportion of standing passengers to seats available, and transfers between buses. This advancement allowed for network improvements to be predicted and compared to base networks.

Many studies exist today that manipulate base models for public transportation in order to increase commuter satisfaction. Berry et. al [1] developed a model of service quality whereby gaps in service delivery were correlated to commuter satisfaction rate. They studied three important discrepancies in the eyes of commuters; the difference between customer expectations and the perception of those expectations according to management, the difference between the perception of expectations and service quality promises, and the difference between those promises and the services that were actually delivered upon commuting. Following the identification of these gaps, a procedure was developed to measure commuters' assessment of quality using the research instrument SERVQUAL. Following this work, Lai and Chen [2] studied the interaction between customer satisfaction and the likelihood of using public transportation in the future. Additionally, work has been done on implementing policies in order to increase public transportation use. Emami et al. [11] used a game theoretic approach in which the competition between public transit and private car is modelled upon the enactment of seven policies meant to incentivise commuters to take public transit.

Our work takes a similar point of view with respect to introducing different policies in attempts to make public transit more attractive. However, our approach includes the

use of agent-based modelling to study the adoptions of different policies on an individual level.

The use of agent-based modelling has opened a new realm of characterizing consumer choices in terms of public transit. The choice of mode of transportation is far more complex than cost and availability of transportation [8]. The works of Hager et al. [8] highlight the importance of individualized characteristics when modelling choice. Other works include data assimilation (DA) to increase the accuracy of model-based predictions, such as the work of Kieu et al. [12]. DA research is motivated largely by atmospheric and oceanographic sciences and has created many significant algorithmic innovations [9]. DA makes an initial forecast (or first guess) and corrects the forecast based on observed data [9]. This process is important to our model as we aim to integrate real data into an agent-based model. Eboli and Mazzulla [6] proposed an index based on consumer satisfaction. Based in Europe, this study was able to quantify the opinions of commuters. They separated consumers based on socio-economic factors (gender, age, employment etc.) and studied how the importance weighting of service aspects changed based on said factors. Their generated index allows for the identification of the problems that cause customer dissatisfaction in terms of public transit qualities. In our use of this consumer satisfaction index, we alter weightings based on income class to build the mathematical framework for our model.

There has been little work done on the combination of agent-based modelling and policy implementation. In this work, we investigate the affect public transportation policies have on individuals from different income classes. Our agents are divided into three subgroups: low income ($0-$49,999), middle income ($50,000-$99,999), and high income ($100,000-$150,000). As mentioned above, we assume each income class gives different weights on factors like time spent commuting, cost of commuting, and comfort level based on mode of transportation. We consider the population in Hamilton, Ontario that commutes to Union Station–located in Toronto, Ontario. Our model assumes an agent has two options to consider each morning: taking public transit (PT) or taking a private car (PC). The aforementioned policies will attempt to increase the attractiveness of PT therefore causing a percentage of the population to switch their habits of taking PC to work, to now adopting PT as their primary mode of transportation.

The motivation behind creating a model that incentivises public transit is to form a better understanding of the operation and consumer dynamics at play during the commuting interval of Ontarians. The idea of optimizing public transit has not been approached the way this study does. We not only study the affects of policy implementation on a population, we also investigate the individual behaviours of agents belonging to different socio-economic classes.

The main objective of this paper is to study the adoption rate by agents of different economic status of public transportation due to policy implementation. The daily changes in agents' decisions to take PT or PC are studied under the influence of their satisfaction from their chosen mode of transportation the previous day. A positive experience while taking PT will increase the chances of an agent taking public transit the following day, whereas a negative experience will decrease the chances of the agent choosing to take PT again. We hypothesize that the implementation of all three policies will increase the percentage of the population that take public transit. Additionally, we hypothesize our last, hybrid, policy will have the biggest affect on commuters' choice to take public transportation

The structure of the paper is as follows:

* Section 2, *Methodology*, will discuss
  (i) Assumptions that are being considered
  (ii) The formulation of the model
  (iii) Which policies will be used in the simulation
  (iv) The simulation and how it works
  (v) Where we collected the data from and how it will be used
* Section 3, *Results*, will discuss
  (i) Our Hypothesis
  (ii) Experimental Design
  (iii) The Base Case
  (iv) Experiment 1
  (v) Experiment 2
  (vi) Experiment 3
  (vii) Overall Results
* Section 4, *Conclusions and Future Work*, will discuss
  (i) Conclusions
  (ii) Future Work and Ideas

2 Methodology

2.1 Assumptions of the Model

There are necessary assumptions that need to be considered upon formulation of the model.

1. Utility maximization: each agent chooses the option that obtains the highest level of satisfaction.
2. Complete information: agents will know about implemented policies from the beginning of the simulation.
3. Static environment: no exterior policies or environmental factors exist in this model.
4. Each agent takes one transportation method per day. If an agent takes public transit to work, we assume they take public transit home.
5. Agents work 5 day, in-person work weeks. This code runs for 365 days, which simulates 73 work weeks.
6. Every agent pays the same public transportation fare, regardless of income level.
7. The only options of transportation are public transit or private car. Options like ride-share, taxi, or carpool are not considered.

8. The commuting population remains constant.

The above assumptions are simplifications of the real-world transportation system. Specifically, assuming agents have knowledge of the policies as soon as they are implemented is a gross simplification of the real-world. This simplification can be addressed in further extensions of the model whereby agents must become aware of policies throughout the simulation.

2.2 Model Formulation

This study develops a probabilistic model in which the daily probability of taking public transit is updated each day. Stopher [14] derived a function in which the choice between PT and PC is expressed in terms of cost and time of journey. Our function uses this idea and also considers a term for satisfaction of the journey. The satisfaction level of a commuter also accounts for the comfort of a commuter in their choice of transit. The weights of this function are calculated using Eboli and Mazzulla's [6] consumer satisfaction index as a base. Our model amalgamates the ideas of [6],[14] into a single probability function. The function is as follows:

$$p_i = -a(d_2 - d_1) - b(t_2 - t_1) + c(s_2 - s_1) \tag{1}$$

The following defines each of the parameters and variables that are contained in the probability model.

Parameters	Notation	Values/Range
Cost of private car	d_1	\$27.50
Cost of public transit	d_2	\$11.50
Time spent commuting in private car	t_2	60 minutes
Time spend commuting in public transit	t_1	85 minutes
Variables		
Importance weight of cost	a	0.2-0.45
Importance weight of time	b	0.36-0.45
Importance weight of satisfaction	c	0.2-0.35
Satisfaction level of public transit	s_2	0-1
Satisfaction level of private car	s_1	0-1

Table 1: Parameters and variables used in the calculation of probability that agent takes PT

As mentioned above, Eboli and Mazzulla's [6] consumer satisfaction index measurement was utilized to obtain values for the variables. By combining aspects of time, cost, and comfort level, these values/ranges were able to be determined. The model presented [6] has service qualities like air conditioning, cleanliness of buses, and environmental impact. This is likely due to the fact that this study was based in Europe, meaning service quality is different from that seen in Canada.

The probability function is dependant on income level. We consider the likelihood that lower income individuals value saving money more than higher income levels, are

less particular about time spent travelling, and value comfort level lower than the higher earning bracket. The importance weights in [6] were used to determine the a, b, and c values for the middle-income individuals. How these weights will be altered in the simulation can be found in section 2.4.

The parameters are derived from GO Transit's web page [15]. Through this data, it was determined that the average cost per bus ride from Hamilton to Toronto is approximately $11.50. We assume an agent must take one transportation mode per day, so the daily cost of PT is $23.00. To calculate the daily cost of owning a private car, we considered 5 factors: insurance, gas, maintenance, car administration fees, and parking fees [5]. It must be noted that the daily cost of owning a private car will vary depending on its age, make, and model. To account for this, we took the average cost of owning a car which was calculated to be $27.50 daily. Using [15], time spent commuting was also obtained. An agent who takes public transport will have approximately an 85 minute one-way commute, whereas one who takes a private car will have a 60 minute one-way commute.

To account for preferences changes, a satisfaction level was included. In this model, s_i takes on a value from 0-1, in which the higher the value, the more satisfied the agent was with his/her choice of transportation method for the day.

The above p-function yields either a positive or negative value. Given a positive value, the agent will choose to take public transit the next day as he/she had a positive experience the day prior. A negative output value deters the agent from choosing public transit the next day, meaning a private car will be taken to and from work. It must also be noted that a value of 0 is possible within this model. In this case, the simulation provides a random result (PT or PC) with equal odds. It is important to note that the magnitude of the values does not matter, only the sign.

2.3 Policies

In order to decide what policies to implement, we used ideas from [11] and a survey. This study summarized the effect that hybrid policies had on the percentage of the population that chose to take public transit. A survey was also conducted, in which 7 questions were asked, one of which was "what would make you more likely to take public transit?". This survey had 175 respondents, all of which were part of the working population or currently enrolled in school. For an in depth explanation of the survey, see the Data Assimilation section. The responses of this survey, coupled with the policies used in [11] were used to create the following three policies.

Policy 1: Increased bus frequency

The first policy considered is an increased frequency of buses. Before the implementation of this policy, we assume buses circulate every hour. By increasing the frequency of the buses, we hypothesize that more commuters will choose to take public transit. This policy introduces the idea that buses will now circulate every 30 minutes. Based on the average time that was calculated to take a bus was 1 hour and 25 minutes, it was then reduced to 1 hour and 10 minutes with this policy to account for a bus that is more often. The increased frequency is hoped to encourage more riders to take buses as this will significantly reduce their waiting time. 27% of survey respondents who suggested ways public transit could be improved said that they would be more likely to take public transit if there was an increased frequency of buses. In addition to convenience and a

reduced travel time, it will also reduce crowding on running buses. This is predicted to increased satisfaction level as comfort mathematically contributes significantly to commuter satisfaction in the model. An additional bus trip every 30 minutes will provide a more lenient commuting schedule which will entice more agents to take a bus to work. The aforementioned solution to the crowding problem is also something respondents of the survey expressed interest in.

Policy 2: Decreased fare for frequent riders

The second policy introduces the idea of fare reduction for frequent riders. This policy creates a sense of loyalty through incrementally decreasing bus fare after a certain number of trips. GO Transit [15] has a policy similar to this, however, we have adjusted the prices to mutually benefit commuters and the transit company. Currently, the first 35 rides of the month cost $11.44 per trip. For rides 36-40, the fare is reduced to $1.54, and free after 40 rides. Basing our altered policy off the notion that agents work 5 day work weeks and must commute to and from work, we wanted commuters to be able to see the affects earlier on in the month. By reducing fare after fewer rides than the current policy, our hope is to establish a sense of loyalty and habit into agents early. The second policy is as follows:

- $11.50 for rides 1-20.
- $8.50 for rides 21-30.
- $5.50 for rides 31+.

Assuming a total of two trips per day for five days per week, commuters will be eligible for a reduced fare at the end of two weeks. This policy aims to encourage occasional bus users to shift their mode of transportation to mainly buses. It will also act as a reward to those who already take the bus daily. This incentive aims to create habits in commuters' mode of transportation. It must also be noted that this policy will reset every 30 days (A month and half). Though this may deter some commuters from public transit at the beginning of each period, it is intended to encourage pattern-building in commuter behaviour while still generating profit to the transit company.

Policy 3: A hybrid model of Policy 1 and Policy 2

The final policy to be implemented is a combination of the two former policies. Since the first policy addresses issues like crowding and bus wait times, and the second policy reduced the cost of taking public transportation each day, we hypothesize that combining them will have the greatest affect on commuters' choice of transportation in favour of public transit. We anticipate this policy will encourage the the highest number of agents of any income class to take public transit as it increases the satisfaction rate at a significant level.

2.4 Simulation

The simulation for this project was coded in Anylogic which is an agent-based modelling software.

There were three different types of agents set up: High Income, Middle Income, and Low Income. These distinguishing factors have all been discussed prior. Each of these agent types had a population of 100. Therefore, while the results would still be the same,

the proportions would be different throughout had we changed the agent population of each group.

Each of these agents were initialized with the following weights, using the probability function that was stated above.

Agent	Satisfaction Weight	Cost Weight	Time Weight
High Income	0.35	0.2	0.45
Middle Income	0.258	0.366	0.376
Low Income	0.2	0.45	0.35

Table 2: Agent sorted by Income Bracket, showing the different weightings that they had been initialized with.

The values for the amount of time spent, cost, and satisfaction rate are described below, they were normalized within the simulation so that the values for time and cost wouldn't skew the equation based on the values that were presented. Satisfaction rates were calculated within the simulation itself. For someone taking a bus, then it was assumed they were more likely to take it again. To start, the satisfaction for the bus was randomized between 0.45-0.50, and the satisfaction for the car was between 0.5-0.55. Whenever an agent would take a bus from there, their satisfaction would increase by 0.002 for the bus, and the satisfaction for the car would decrease by 0.002. This operation was reversed if a car was chosen. While this was a minimal increase/decrease, it showed a slight increase in willingness to take a specific vehicle again.

Variables	Values
Bus Time	0.725
Car Time	0.50
Bus Ticket Cost	0.115
Car Ticket Cost	0.275
Bus Satisfaction Rate	0.45-0.50
Bus Satisfaction Rate	0.50-0.55

Table 3: Variables used within the code with their normalized values.

Within the simulation, there were also events that were added in that would simulate real-life scenarios. The first two events that were added in was an occurrence of traffic for a car, and an occurrence of traffic for the bus. In these scenarios, they were scheduled to go off at different times within the simulation, and occur once every 5 days. They shouldn't have an overlap because of this. For both instances, the variable for either bus time or car time was increased by 0.5, to simulate another hour of traffic being added on. There was also an event added on to simulate running late, this would be added once every 7 days, and it would change the time weighting for each of the agents. It would change the weighting differently depending on their income bracket. The main reason this was done is because all of the agents value these differently, so it wouldn't make sense to change them at the same rate. For low income, the time-weighting increases by 0.1,

for middle income it increases by 0.15 and for high income it increases by 0.2. After these events, the values are all reset to ensure it only happened for that specific day. It should be noted that the agents see the changes in events before they make their decisions.

The agents are given memory before the simulation starts, and throughout the simulation also. The agents are given randomized memory from before the simulation start with specific ranges for income class. This would simulate the choices that were made the days before from the probability function.

1. **High Income Class** has a value that is randomized between -10 to 0. This was given as High Income would be the most adverse to taking transit at first.

2. **Middle Income Class** has a value that is randomized between -7.5 to 2.5, and this was given as they are not necessarily opposed to taking public transit, but with access to a car, they are not likely to start off with taking transit without incentive.

3. **Low Income Class** has a value that is randomized between -5 to 5 and this is because with presumed access to a car, then they are equally likely to take both at the start.

Within the probability function, there was an added component for memory. Memory is only stored for the last ten days of the simulation, and how much it impacts their choices varies from the length of time of the present day to that decision in the simulation.

The updated function is as follows:

$$p_i = -a(d_2 - d_1) - b(t_2 - t_1) + c(s_2 - s_1) + \sum_{n=1}^{10} \delta^{n+1} m_n \tag{2}$$

1. m represents the memory which is the value of the probability function from a specific day day

2. n represents the number of days this was before the present day in the simulation

3. δ represents a randomized value between 0 and 1 for each agent

2.5 Data Assimilation

The data used in this paper came from various sources. As previously mentioned, the different weightings for each income class comes from [6]. Eboli and Mazzulla considered an experimental case study in which the transit services in a mid-sized urban area located in Italy were studied. Habitual commuters of two bus lines were issued a survey that collected 6 characteristics: gender, age, employment, sector of employment, professional position, and family income level. Next, commuters were asked to rate the importance and satisfaction of 26 service attributes. The study revealed that the service attributes that contribute most significantly to commuter satisfaction are ease of purchasing ticket, security on buses, reliability of buses, and personnel appearance.

Data was mainly collected from census records or the City of Hamilton Data Catalogue [4]. The census data collected served as our population statistics, specifically, the main mode of commuting by commuting duration, time leaving for work, age and gender. Statistics Canada [3] helped identify the number of agents and size of population that each agent represented. By amalgamating the number of respondents who commuted for

45 to 59 minutes via public transit (2360) and the number of respondents who commuted 60 minutes and over via public transit (4890), the estimated count of individuals affected by the research done in this paper was 7250 people. Approximately 7% of the population of Hamilton.

This research focuses around three distinct agent cohorts that were divided based on the income bracket they fell under. This meant there would be a cohort of agents representing a low, medium, and high-income bracket. To determine the thresholds for grouping agents based on this criteria, it was decided to use the Census Income data product [4] available in the City of Hamilton Data Catalogue. The Census Income data product logged counts of the number of individuals in each income threshold specified. These thresholds were distinguished in $25 000 increments, for example, households with income under $25 000 and households with income between 25 and 49 000 dollars are two different counts.

Given the research was focused around 3 income cohorts, the income data was amalgamated to represent 3 income classes: 0-49k, 50-99k and 100-149k. By amalgamating these counts, the fractions of each agent cohort that made up the 7250 members of the population was determined. The low income agent cohort (0-49K) would represent 33% of the population (2393 individuals), the medium income agent cohort (50-99K) would represent 41% of the population (2973 individuals) and the high income agent cohort would represent 26% of the population (1885 individuals).

Finally, the data collected on the current rate of public transit adoption was derived from the Statistics Canada data product Public transit in a post-COVID-19 Canada. This product states the proportion of pre-pandemic transit passengers in Ontario as 25.6% of the average Census Metropolitan Area (CMA). This is used in this research as a benchmark for the proportion of Hamilton that are established as public transit adopters. The initial adoption fraction in the simulation was set to 25% and the experiment results are in comparison with this statistic.

To account for the characteristics of commuters of different countries, we conducted a survey to see what attributes were considered most valuable in a Canadian setting. The survey consisted of 7 questions pertaining to demographic and preferences of respondents. Data collected includes age, possession of valid driver's license, number of people in household, number of vehicles per household, average trips away from home per day, method of transportation, and what factors would influence respondents to take public transit. From this data, it was seen that only 36.6% of respondents take public transit, while the vast majority (82.9%) take private cars, as seen in Figure 1. The final question asked was "what would make you more likely to take public transit?" in which 153 out of 175 respondents answered. Of the 153 responses, 24.8% of people said that there is nothing that would incentivise them to take public transit. The other responses consisted of factors like lower cost, more frequent buses, safety, and comfort. From this information, we were able to construct the three policies in attempts to increase public transit usage.

Survey Question	Response	Proportion
Age?	Under 18	16.6%
	18-24	65.7%
	25-34	2.3%
	35-44	4%
	45-54	8.6%
	55-64	1.7%
	65+	1.1%
Possession of valid driver's license?	Yes	83.3%
	No	16.7%
How many people reside in your household?	1	4%
	2	10.3%
	3	17.7%
	4	42.9%
	5	17.1%
	6+	8%
How many vehicles does your household own?	0	4%
	1	13.7%
	2	46.3%
	3	22.9%
	4+	13.1%
How many daily trips do you make?	0-2	53.1%
	3-5	43.4%
	6-8	1.1%
	9+	2.3%
How are you most likely to get around town?	City bus	36.6%
	Private transit	82.9%
	Ride-share	16%
	Bike	9.1%
	Train/Subway	6.3%
	Walking	38.3%
	School bus	9.7%
	Other	3%

Table 4: Google Docs survey results of 175 respondents

3 Results

3.1 Hypothesis

Before looking at the results, the hypothesis should be reviewed and discussed. We hypothesize that upon the implementation of Policy 3, the greatest amount of commuters will adopt the use of public transit. Policy 3 is able to dissolve the most issues that the commuters are posed with. In addition, we hypothesize that Policy 1 will be the least effective since the bus time would stay consistent, whereas, a user taking the bus more often would result in even cheaper fares. While these weightings were almost equally distributed, the cheaper fares would have better benefits. Despite Policy 1 decreasing

wait time, it will still not be as efficient as an agent taking their own car. For this reason, it will likely encourage the least amount of people to take public transit.

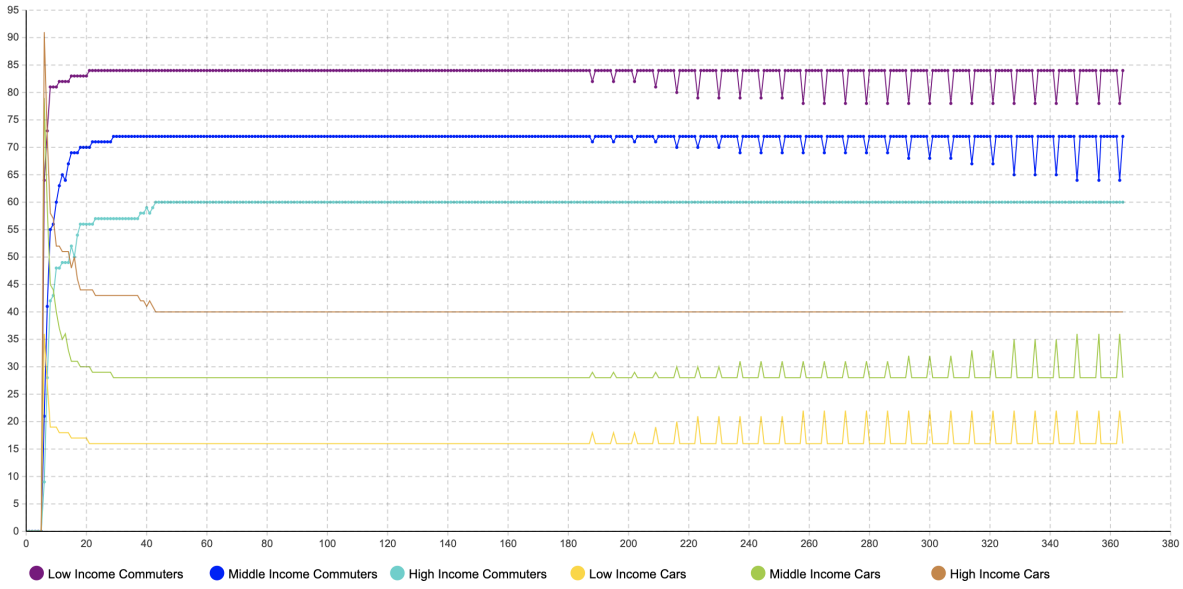
3.2 Experimental Design

To test the efficacy of each policy, we ran four experiments. Each experiment was run 10 times for 365 days. In order to compare the policy implementation experiments, graphs of commuter behaviour were generated. Graphs of the same experiment produced similar behaviour, so a graph of each experiment is included in the following sections from a random run.

We aim to study the amount of commuters who take public transit by the end of the simulation once a policy has been implemented. We compare these numbers to the case in which no policies are active, meaning no one has any external incentive to take public transit. At the end of each run, the number of agents who take public transit on day 365 is recorded from each income class. We use these numbers to generate an average between the 10 runs. These proportions will allow for the comparison of policy efficacy.

3.3 Base Case

First, the simulation will be run without the implementation of any policies. This will allow for a comparison to happen not just between polices, but also to the base case to see how effective each policy was in comparison. The following graph shows which choice of vehicle was made by each agent in the differing income classes in our first run of the simulation. This graph was used as all of the graphs show similar behaviour.



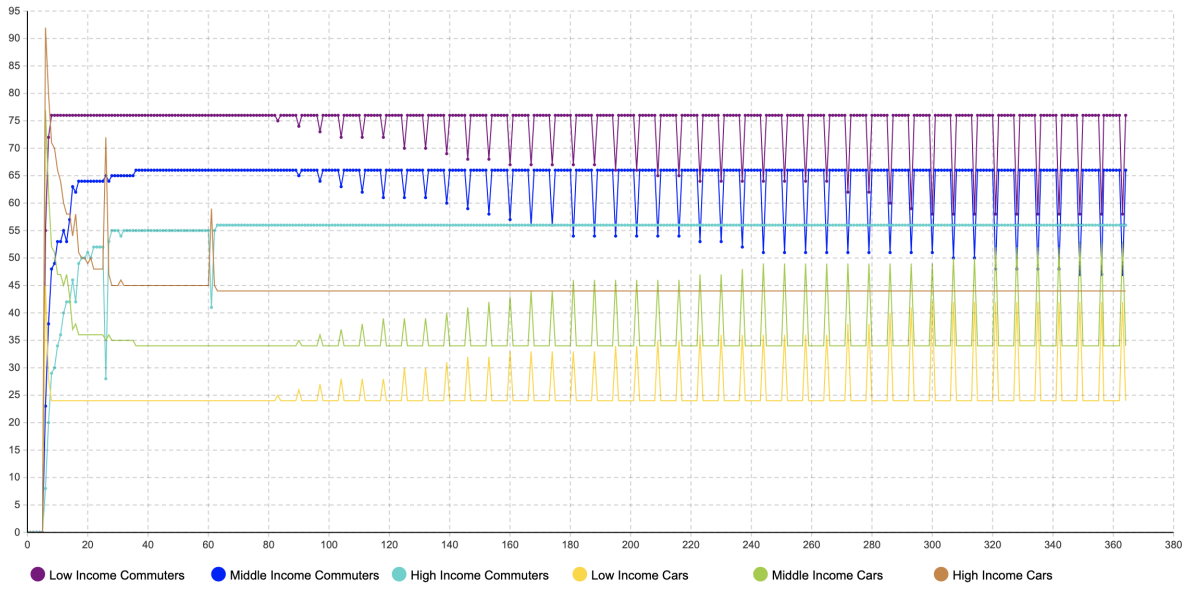
Choices of Vehicle by Income per day for Run 1

At the beginning of the simulation, each of the income classes acted in terms of taking public transit or car to commute to work. These values were randomized at the beginning of the simulation, and were discussed in the section above. From day 45 to approximately

day 190, we can see that no agents change between taking public transit to car or vice versa for the entire time period. Then, just after day 190, we can see a small movement begin, which grows with time. This graph begins to represent a sinusoidal function. This was an interesting result as it began to change behaviour after approximately 190 days of being stagnant. We deduced that this was due to the agents being able to observe potential traffic or bus delays that could cause them to be late for work, so on those days they would take their own car. On those specific days, they are able to avoid being late by taking their own vehicle. We notice the high income class doesn't change at all. This may be attributed to the fact that their p-function has different weightings for time, satisfaction, and cost. The different weights cause the high income agents to have p-values that do not require changing their decision, even if they are faced with delays.

3.4 Experiment 1: Policy 1

For these trials, we implemented the policy of increasing the frequency of buses. This should increase the number of agents taking public transit since it will increase their satisfaction rate due to a decreased wait time. The following graph shows the results from run 2 of our simulations:



Choices of Vehicle by Income per day for Run 2

Analyzing this graph, we see a result that is the opposite of what we hypothesized. Here, we see a larger change in amplitude than in the base case. Additionally, we observe that the fluctuations in choices occur much earlier in the simulation. This policy had a negative affect on the proportion of low- and middle-income bus users as less agents took public transit after this implementation. However, the proportion of high-income agents who were frequently taking the bus after 365 days increased by almost 6%. This result is surprising as we did not expect to see policy implementation have any significant affect on high-income commuters. We see that we also have a big spike right at the beginning of the simulation, and this occurs only in Experiment 2 and Experiment 4. We can see that the spike was specifically due to this policy measure.

3.5 Experiment 2: Policy 2

For experiment 2, the policy being implemented is a decrease fare for frequent riders. Just as a reminder, this means that for rides 1-20 that the agents take, it will cost \$11.50, for rides 21-30 it will cost \$8.50, and for rides 31+ it will cost \$5.50. These rides represent a single one-way trip. Therefore, if an agent decides to take transit for the day, then they will have done 2 trips. This should increase the number of low and middle income class commuters that use public transit since the cost is more important to them. The following graph shows the results of run 1 when this policy was implemented:

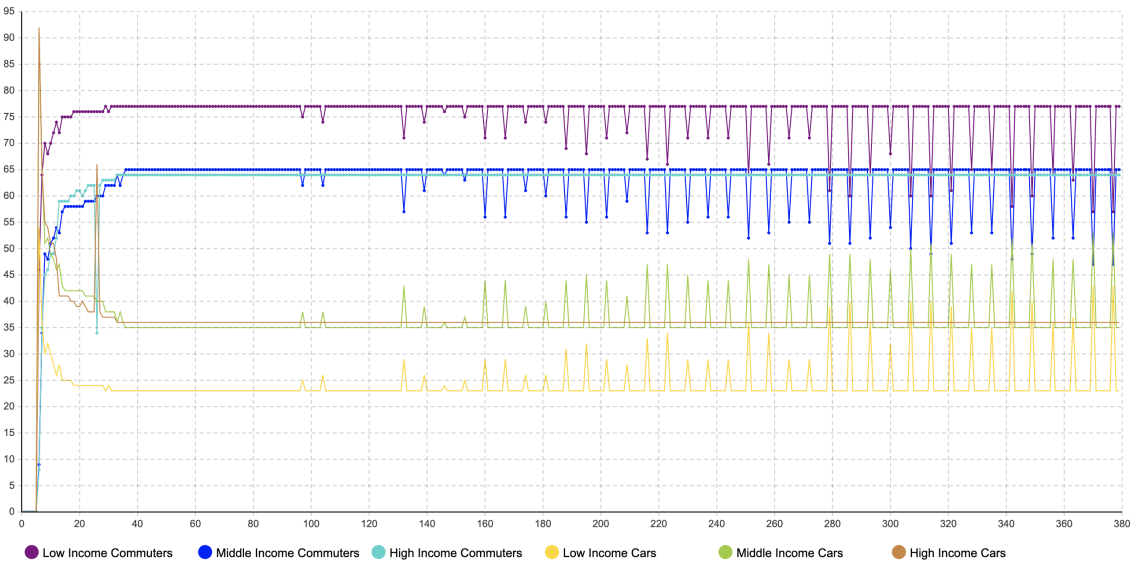


Choices of Vehicle by Income per day for Run 1

Here, we see the drastic change in ridership begins around day 160, and that they were larger than in the base case. The larger amplitude would make sense in theory because more people would be taking public transit, but since the number of public transit users decreased, this again is an illogical outcome. Once again, compared to no policy, the amount of public transit users decreases, and compared to Policy 1, we see a larger amount of public transit users for the lower and higher income classes, while there is a decreased amount for the middle income class.

3.6 Experiment 3: Policy 3

In Experiment 3, the policy that is implemented is a combination of increased bus frequency and decreased bus fare for frequent rides. It is expected that the combination of policies will lead to the largest number of agents taking public transit since it benefits the greatest percentage. The following graph depicts run 10 of when this policy was implemented:



Choices of Vehicle by Income per day for Run 10

This was the lowest amount of riders for public transit that was found, and the changes started very early on. While it still ended up that there were more users taking transit than taking a car, the results seem unlikely based on what has been hypothesized. We again see that when compared to the base case, less agents overall took public transit. This is an illogical outcome since incentivising the agents, had a greater population stop taking public transit, even though it benefits them more than in the initial case.

3.7 Overall Results

Proportion of bus users	Base Case	Policy 1	Policy 2	Policy 3
Low-income	85.8%	82.4%	85.6%	83.1%
Middle-income	64.9%	63.6%	62.8%	63.2%
High-income	63.3%	69.1%	64.8%	65.9%

Table 5: Average proportion (%) of bus users at day 365 of simulation

Looking at Table 5, we can see that the overall percentages of the population who took the bus after 1 year changed drastically depending on the policy and income bracket.

In terms of low-income bracket, the best performance was found in the base case, but Policy 2, which was the reduced fairs was quite close in comparison. This logically made sense, as the lowest income bracket had the highest weighting for the total cost.

In terms of the middle-income, the best performance was in the base case, which was surprising given the different polices, and all of the values were within 2.1% of each other, which was the bracket that showed the least movement. This was the most surprising, as this was expected to be a bracket that showed the most movement. Policy 2 was the least impactful policy, and based on the weightings for each of the different variables, it seems to line up that this would be a policy that didn't matter as much for this bracket.

The highest-income bracket showed drastic differences depending on the simulation, and had its best result with Policy 1 which was for increased bus service. The highest

weighting this bracket had was for time, so this result followed closely to the anticipated change in the movement of the high-income. The lowest value was the base case, so the policies were the most effective for this bracket.

For each of the experiments, with the fluctuations found, we attribute that to a small group in both middle-income and low-income who notice that heavy bus traffic occurs once every 5 days, and take their car only on those days, but are otherwise loyal bus riders.

4 Conclusions and Future Work

4.1 Conclusions

After viewing the results of the experiments, the hypothesis should be revisited in order to look at its accuracy. It was expected that all policies would have a positive affect on the amount of commuters who choose to take public transit. We hypothesized that Policy 3 would have the largest impact on proportion size as it combines both of the prior policies. This was not the case as it lowered the amount of low- and middle-income agents who take the bus, while slightly increasing the amount of high-income agents who take the bus. Additionally, we hypothesized that Policy 1 would be the least effective, this was not true as it had the largest impact on high-income agents, increasing the proportion of bus users by 5.8%.

It is observed that the high-income class is most receptive to the implementation of different policies, and tends to remain static after 80 days of simulating. Unlike the other two income classes, the high-income class does not fluctuate between choosing PC and PT every 5 days.

Revisiting the probability function would also help to understand where the skewed results came from. Realistically, it would make sense that once certain values within had been changed, the number of commuters should've increased. Potentially adding in more categories would've made the function more accurate.

Overall, the base case made sense to a certain extent with the average number of commuters. However, the drastic increase in commuters at the beginning of the base case doesn't align with our expected values for each income bracket. Adding memory in the beginning with the expected values for the income brackets helped with the initial population, but it still had a steep increase regardless. Therefore, we want to revisit our probability function to understand why our function increased how it did.

4.2 Future work and ideas

There are many ways that this study could be progressed, and due to the period of time for this research, we weren't able to implement these ideas. However, for potential future work on this topic, there are a few changes we would like to make.

1. Remove static environment assumption

    - In a real-world situation, we know that there is always competition. For this reason, to advance this study in the future, other transit companies should be considered which could also release their own personalized policies. Simulating a competition between multiple companies would be interesting to see how the original policies are affected while others are implemented at specified points

in time. This could also allow the study to look at the possibility of new routes that agents could take.

2. Different policies

   - Realistically, there are infinite amounts of policies and policy combinations that could be attempted in a simulation. It would be very interesting to see how different combinations could provide different results, while also keeping the policies realistic. Having a number of routes that busses could take would provide different options for the agents.

3. Ride sharing and other transportation options

   - Introducing other ways that commuters can make it to and from work would add a level of realism into the study since realistically there are multiple methods other than a car or bus that people take to work. A good way to incorporate them would be to include all people in a range of the starting point. For example, the study could include a 10km range from the route that the bus takes to get from Hamilton to Union Station. This means that it could include people from the entire duration of the trip, rather than just the start. This would increase the number of people that walk and bike since their distance to work is very short. On top of all these points, ride sharing is an up and coming service which could continue to grow in the near future. Due to the low cost and convenience of ride sharing, it will likely sway a portion of commuters away from public transit and influence them towards taking ride sharing. Including this in the study would likely yield a very interesting result.

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